AQI Prediction & Forecasting

Chennai — SARIMA Models + CPCB India Sub-Index

Project Report

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1. Abstract

This project delivers an end-to-end air quality forecasting and assessment tool for the city of **Chennai, India**. Using historical pollutant concentration data from the Central Pollution Control Board (CPCB), seven independent **SARIMA** (Seasonal Autoregressive Integrated Moving Average) models were trained — one per pollutant ($PM_{2.5}$, PM_{10}, NO_2, SO_2, O_3, CO) and one for the overall Air Quality Index. Monthly pollutant forecasts are fed into the official **Indian CPCB sub-index formula** to derive a colour-coded, health-advice-annotated AQI. The original Tkinter prototype was rewritten as a polished **Streamlit web application** with two modes of use: forecasting by month, and live AQI calculation from user-supplied concentrations.

2. Introduction

Air pollution is a primary driver of respiratory and cardiovascular disease in Indian metropolitan areas, and Chennai — with its mix of industrial activity, heavy vehicular load, and coastal meteorology — reports sustained exceedances of CPCB safe limits across multiple pollutants. Reliable short- to medium-horizon forecasts support: (1) public-health advisories, (2) traffic and industrial-emission planning, and (3) individual awareness during high-risk periods.

This project combines **classical time-series modelling** (SARIMA) with the **official Indian CPCB AQI methodology** to produce a transparent, explainable, and reproducible forecasting tool. Unlike opaque black-box models, every number in the output can be traced back to a concrete pollutant concentration forecast and an explicit piecewise-linear sub-index formula published by CPCB.

2.1 Objectives

- Forecast monthly concentrations of six key pollutants in Chennai using SARIMA.
- Convert pollutant forecasts into the official CPCB overall AQI via the sub-index formula.
- Provide a live AQI calculator for user-supplied observed concentrations.
- Display colour-coded severity and actionable health advice across six CPCB categories.
- Replace the original Tkinter prototype with a modern web-based interface.

3. Data

The dataset is the Kaggle *Air Quality Data in India (2015–2020)* release, compiled from CPCB continuous monitoring stations across Indian cities. Only the **Chennai** subset was used; daily station-level observations were aggregated to **monthly means** before SARIMA fitting.

File	Description	Scope
city_day.csv	Daily pollutant + AQI per city	All cities
station_day.csv	Daily pollutant + AQI per station	All stations
stations.csv	Station metadata (ID, city, type)	All stations

Chennai monitoring stations in this dataset include *Alandur Bus Depot*, *Manali*, *Manali Village*, *Velachery Res. Area*, among others. Measurements are reported in µg/m^3 for particulates and gases, with the single exception of CO which is reported in mg/m^3.

4. Methodology

4.1 Why SARIMA?

Monthly pollutant concentrations in Chennai exhibit a strong **annual seasonal cycle** driven by the northeast monsoon (Oct–Dec), summer thermal inversions, and festival-linked emission spikes (Nov, Jan). A SARIMA model captures three components simultaneously:

- **Autoregressive (AR)** component — persistence from recent months.
- **Integrated (I)** component — differencing to remove trend and achieve stationarity.
- **Moving average (MA)** component — residual correction from past forecast errors.
- Plus a **seasonal block (P, D, Q, s)** — the same structure applied at lag s months to capture the annual cycle.

The model order (p, d, q)(P, D, Q, s) was selected per pollutant through (i) an Augmented Dickey-Fuller stationarity check, (ii) ACF/PACF inspection, and (iii) AIC-minimising grid search. Training procedure is documented in the per-pollutant notebooks under notebooks/training/.

4.2 CPCB Sub-Index Formula

The Indian CPCB National AQI is a **piecewise-linear transform** from pollutant concentration to a 0–500 sub-index. Each pollutant has six concentration bands (Good, Satisfactory, Moderate, Poor, Very Poor, Severe). For a concentration C_p falling into the band with break-points BP_{LO}, BP_{HI} and index limits I_{LO}, I_{HI}:

```
I_HI – I_LO
I_p = ———————— · ( C_p – BP_LO ) + I_LO
BP_HI – BP_LO

Overall AQI = max( I_p ) across all pollutants
```

The **maximum sub-index rule** means the single worst pollutant drives the overall score; the application surfaces this as the *dominant pollutant* readout so users understand which component is actually responsible for the rating.

5. Model Details

Seven SARIMA models were fit using statsmodels.tsa.statespace.sarimax. The table below summarises the fitted specification per pollutant.

Target	Order (p,d,q)	Seasonal (P,D,Q,s)	Obs.	AIC
Overall AQI	(0, 0, 1)	(1, 0, 1, 24)	65	714.5
O_3 (Ozone)	(2, 1, 3)	(1, 0, 1, 12)	65	382.4

Target	Order (p,d,q)	Seasonal (P,D,Q,s)	Obs.	AIC
PM2.5	(2, 0, 0)	(1, 0, 1, 12)	41	368.0
PM10	(2, 0, 0)	(1, 0, 1, 12)	41	429.8
NO_2	(2, 1, 3)	(1, 0, 1, 12)	41	257.6
SO_2	(3, 1, 3)	(1, 0, 1, 12)	41	240.2
CO	(5, 0, 1)	(1, 0, 1, 12)	41	164.4

Training ranges: Overall AQI and O_3 were trained on Jan 2015 — May 2020 (65 monthly observations). The remaining pollutants were trained on Jan 2015 — May 2018 (41 monthly observations), reflecting the data availability of each sensor at the Chennai stations.

Caveat. SARIMA extrapolates the seasonal pattern observed during training. Forecasts far beyond the training horizon rely on the assumption that this seasonal structure persists; long-horizon outputs should therefore be treated as *indicative*, not as exact point forecasts.

6. CPCB Breakpoint Grid

The full breakpoint table used by the application is reproduced below. All concentrations are in µg/m^3 except CO (mg/m^3). The same six index ranges (0–50, 51–100, 101–200, 201–300, 301–400, 401–500) apply to every pollutant; only the concentration boundaries vary.

Category (Index)	PM2.5	PM10	NO_2	SO_2	O_3	CO
Good (0–50)	0–30	0–50	0–40	0–40	0–50	0–1.0
Satisfactory (51–100)	31–60	51–100	41–80	41–80	51–100	1.1–2.0
Moderate (101–200)	61–90	101–250	81–180	81–380	101–168	2.1–10
Poor (201–300)	91–120	251–350	181–280	381–800	169–208	11–17
Very Poor (301–400)	121–250	351–430	281–400	801–1600	209–748	18–34
Severe (401–500)	251+	431+	401+	1601+	749+	35+

7. Application Architecture

The application is a single-file Streamlit app (app.py) organised around two tabs. The original Tkinter GUIs (kept for reference in notebooks/legacy_gui/) are replaced by a single web front-end.

7.1 Forecast by Month Tab

- Date picker selects the forecast target month.
- Each pollutant SARIMA produces a concentration forecast at that horizon.
- Per-pollutant CPCB sub-index is computed from the forecast.

- Overall AQI = max of the six sub-indices.
- A standalone AQI SARIMA prediction is shown alongside as a cross-check.
- Colour-coded severity banner with six CPCB categories and health advice.
- One-click CSV export of the per-month forecast.

7.2 AQI Calculator Tab

- Six numeric inputs: PM2.5, PM10, NO_2, SO_2, O_3 (µg/m^3), CO (mg/m^3).
- Per-pollutant sub-index table updates live as values change.
- Overall AQI and dominant pollutant are recomputed on every keystroke.
- Colour-coded banner + health advice mirror the forecast tab.
- Expandable reference table of the full CPCB breakpoint grid.

8. Application Screenshots

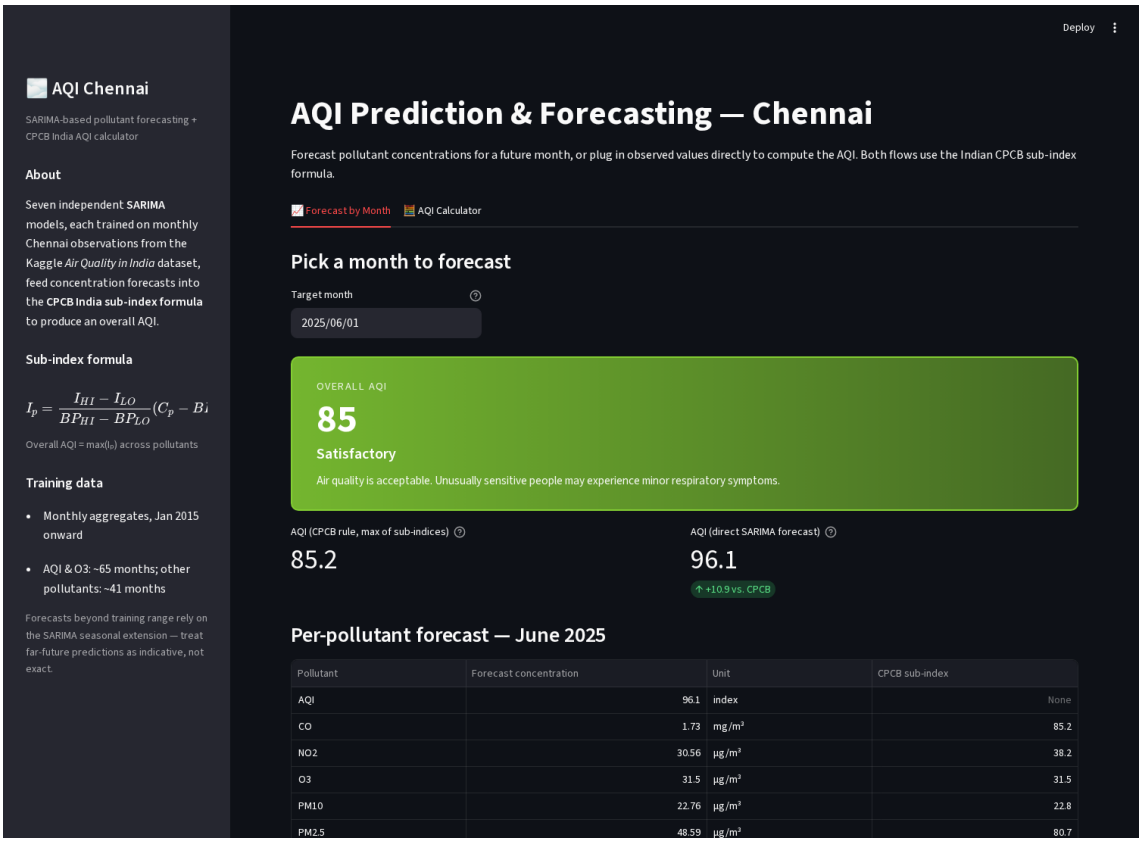

Figure 1 — Forecast by Month tab: month picker, CPCB vs. SARIMA AQI side-by-side, per-pollutant forecast table with units.

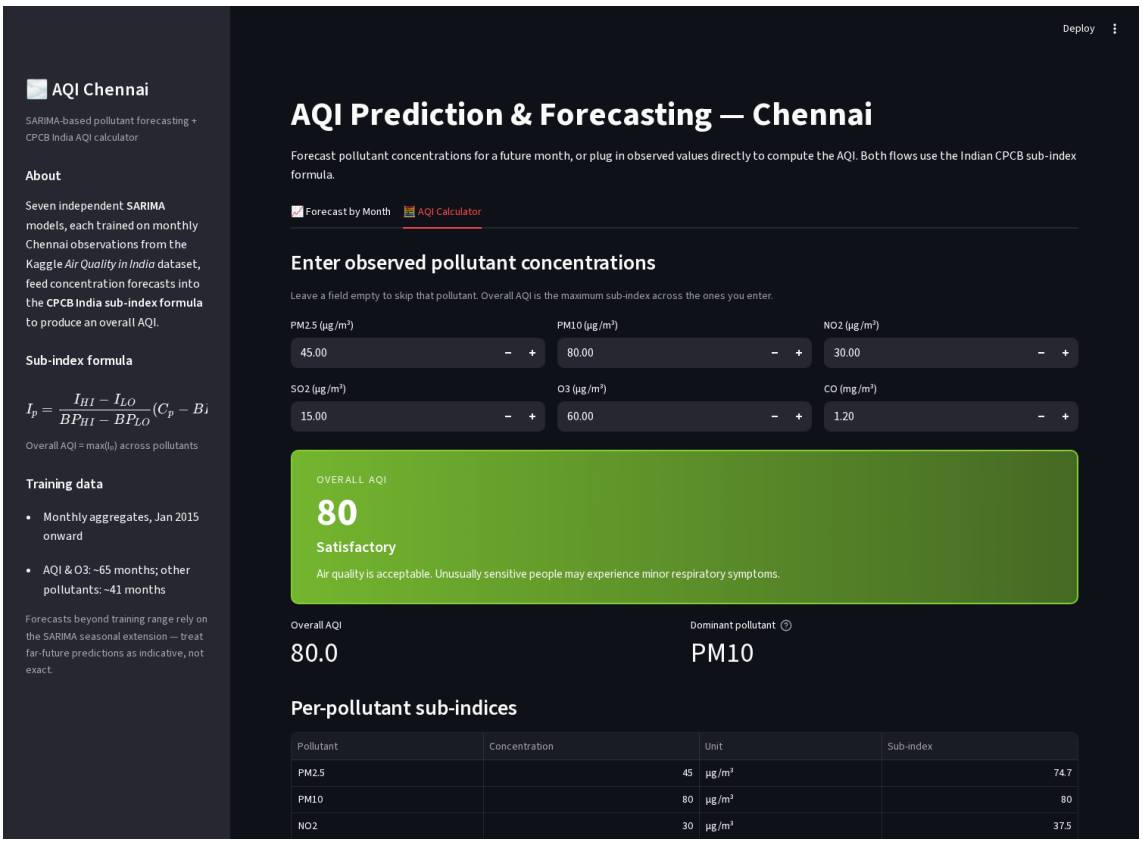

Figure 2 — AQI Calculator tab: six numeric inputs, live colour-coded AQI banner, dominant-pollutant readout, per-pollutant sub-index table.

9. Repository Layout

```
.
■■■ app.py # Streamlit app (forecast + calculator)
■■■ requirements.txt # streamlit, statsmodels, pandas, numpy
■■■ models/ # Pickled SARIMAX models
■ ■■■ aqi.pkl co.pkl no2.pkl
■ ■■■ o3.pkl pm10.pkl pm25.pkl so2.pkl
■■■ data/ # Kaggle CPCB CSVs
■ ■■■ city_day.csv
■ ■■■ station_day.csv
■ ■■■ stations.csv
■■■ notebooks/
■ ■■■ training/ # One fit notebook per pollutant
■ ■■■ legacy_gui/ # Original Tkinter GUIs (archived)
■■■ docs/
■■■ images/ # Screenshots used in README + report
```

10. Running the Application

Install dependencies and launch Streamlit:

```
pip install -r requirements.txt
streamlit run app.py
```

The app opens at http://localhost:8501 in the system browser.

11. Limitations and Future Work

- **Training horizon.** Most pollutant models end in May 2018; re-training on the full 2015–2020 span would improve long-horizon accuracy.
- **Monthly granularity.** The app forecasts monthly means; daily or hourly forecasting would require finer-grained SARIMA variants or deep-learning alternatives (LSTM, Temporal Fusion Transformer).
- **Chennai-only.** The same pipeline generalises to any CPCB city given sufficient station coverage — a city selector is a natural extension.
- **Exogenous regressors.** Meteorology (temperature, humidity, wind speed, rainfall) are known drivers and would naturally fit a SARIMAX extension.
- **Deployment.** The app is currently run locally; hosting on Streamlit Community Cloud would provide a live shareable URL.

12. Conclusion

This project demonstrates that **classical time-series modelling combined with the official CPCB AQI methodology** is sufficient to build a transparent, reproducible, and end-user-friendly air-quality forecasting tool. The rewrite from a Tkinter prototype to a Streamlit web application significantly improves usability while preserving the underlying SARIMA models and CPCB sub-index mathematics. The codebase, models, notebooks, and this report are released under the Apache-2.0

license.

Repository: **github.com/NishanthSundaran/AQI_prediction**

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